Coherent Twinning of a Pearcey Beam in a Frozen Ideal-Lens Geometry: A Bounded Test of a Bilinear Core-Efficiency Law

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Abstract

We test a model-authored hypothesis for coherent twinning of a scalar Pearcey-type input in a frozen ideal-lens geometry. The registered observable is the fixed-core core efficiency $\overline{C}$ after propagation through a thin lens. The hypothesis asserts that the coherent-minus-incoherent change $\Delta\eta$ follows a zero-free-parameter bilinear weight law, that a calibration at one weight predicts held-out weights, that a relative phase of π reverses the sign exactly, and that the coefficient distinguishes a caustic input from a parity-matched non-centrosymmetric control. Using the executed angular-spectrum runs at $n = 192$ and confirmation variants at $n = 256, 384, 576$, an independent Fresnel run, and wavelength and sampling variants, we find a positive weight trend at zero relative phase: the coherent arm raises $\overline{C}$ from 0.518068 to 0.548308 at the calibration point ($\Delta\eta = +0.030240$). Under the inferred mapping $\alpha = 1/(1 + a^2)$, the three held-out exploratory weights give $\Delta\eta = 0.034366$, 0.033484, and 0.025198, with magnitudes of relative deviation 9.1%, 7.1%, and 11.1% from the calibration prediction using $K\Re O = 0.03780$. However, the decisive phase-reversal prediction fails: at the same weight and a relative phase π, $\Delta\eta = -0.123194$, so $|\Delta\eta(\pi) + \Delta\eta(0)|/|\Delta\eta(0)| = 3.07$, far above the pre-registered 0.20 bound. The sign reverses, but the exact magnitude reversal does not. The normalization denominator and the difference between the core-local cross integral and the global overlap make exact negation algebraically implausible for the stated unit-power construction. Caustic specificity is untested because the matched non-centrosymmetric genericity arm was not executed. Optimized classical phase-shaping controls reach $\overline{C} = 0.599$–0.619, above the coherent Pearcey arm, so the twinning effect is not a competitive focusing advantage. We report a bounded negative for the proposed zero-parameter global-overlap law, while retaining the observed coherent cross-term shaping as a numerical effect requiring a core-local model and further controls.

1 Introduction

Structured scalar fields can develop sharp transverse features when a phase singularity or a caustic is focused by a lens. Pearcey-type beams are a useful test case because their transverse structure is non-centrosymmetric and has been studied for autofocusing and self-healing [3], for symmetric Pearcey–Gaussian variants [7], for masked Pearcey arrays [6], and for propagation in anisotropic media [8]. Optical catastrophe beams more generally provide a route to controlled transverse morphologies [9]. The present question is narrower than a new beam-family proposal: in a frozen ideal-lens geometry, does a coherent superposition with the 180° rotated copy produce a fixed-core efficiency change that follows a zero-free-parameter bilinear law?

The hypothesis tested here was formulated by the model before the numerical runs. It asks whether

$$\Delta\eta(\alpha) = K\, 2\sqrt{\alpha(1-\alpha)}\, \Re O \tag{1}$$

holds for the registered observable $\overline{C}$, with O the input overlap between the field and its 180° rotation. It further asks whether K measured at one weight predicts held-out weights, whether a relative phase π flips the sign exactly, and whether K is shared with a parity-matched non-centrosymmetric control. The question is intentionally falsifiable: the phase-reversal case has

a sharp magnitude criterion, and the caustic-specificity claim requires a control arm that can demote the effect to generic coherent cross-term shaping.

We retain the registered observable $\overline{C}$ throughout and do not replace it with an axial-floor metric or a different focusing diagnostic. The central result is that the weight trend is compatible with a bilinear cross-term account at zero relative phase, but the exact phase-reversal prediction fails decisively. The data therefore support neither the proposed zero-parameter global-overlap law nor a caustic-specificity claim. They do support a bounded statement: coherent superposition with an inverted copy changes fixed-core concentration, and the change depends on relative phase and weight in a way that requires the normalization and core-local cross integral to be modeled explicitly.

2 Related Work

Pearcey beams and their Gaussian-apodized forms have been characterized as autofocusing and self-healing fields [3, 7]. Masked Pearcey arrays extend the transverse shaping toolkit [6], and propagation in uniaxial crystals shows how the medium can separate polarization and diffraction effects [8]. Catastrophe-beam studies provide a broader language for caustic morphology [9]. These works motivate the use of a Pearcey-type input as a non-centrosymmetric test field, but they do not supply the present coherent-twinning law or its normalization.

The parity operation used here is related to mode sorting by two-dimensional parity [1]. A 180° rotation implements an inversion operation on the transverse field, so the coherent cross term is a parity-sensitive overlap. Phase-only beam shaping is also relevant as a baseline: optimized phase masks can improve beam quality or concentration without any coherent twin [5]. We therefore treat optimized classical phase-shaping controls as baseline context for the same absolute observable, not as a test of the twinning law and not as a negated optimizer score.

Numerically, we use a scalar angular-spectrum propagator with a conservative radial band limit. The band-limited angular-spectrum method is standard for free-space propagation [2]; the cutoff used here is a conservative radial cutoff, not the complete Matsushima construction. This distinction matters because the present evidence is a numerical survey, not a claim of grid-independent convergence for all observables.

3 Theoretical model

The input field is normalized to power P on the numerical grid:

$$U_0(x_j, y_k) = \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} |V_{ab}|^2}}\, V(x_j, y_k), \tag{2}$$

where V is the selected scalar family. For a Pearcey-type input we use

$$V(x,y) = C_n\Big(\frac{x}{s}, \frac{y}{s}\Big) \exp\!\left(-\frac{x^2+y^2}{w^2}\right), \quad C_n(u,v) = \int \exp\big[i\left(t^n + vt^2 + ut\right)\big]\, dt, \tag{3}$$

with $n = 4$ for Pearcey and $n = 6$ for Butterfly. The executed input is Pearcey with $n = 4$, scale $s = 150\,\mu$m, and Gaussian envelope radius $w = 300\,\mu$m from the optical setup. The Butterfly control amplitude is zero in this study. For completeness, the Airy family used by the registry is

$$V(x,y) = \mathrm{Ai}\Big(\frac{x}{s}\Big)\,\mathrm{Ai}\Big(\frac{y}{s}\Big) \exp\left[\alpha_{\mathrm{ap}}\left(\frac{x+y}{s}\right)\right] \tag{4}$$

inside the aperture; the apodization field affects Airy inputs only and does not enter the Pearcey input used here.

Propagation through the frozen ideal lens and free space is modeled by

$$U_z = \mathcal{F}^{-1}\Big[\mathcal{F}(U_0)\, B_z\, \exp\Big(i2\pi z\sqrt{\lambda^{-2} - f_x^2 - f_y^2}\Big)\Big], \tag{5}$$

where B_z is a conservative radial bandwidth cutoff. The lens phase is applied with focal length $f = 0.06$ m. The composition uses an explicit rotation θ, relative phase δ, and amplitude weight a:

$$U_c = \sqrt{\frac{P}{\int |U + ae^{i\delta}\mathcal{R}_\theta U|^2\, dA}}\, \Big(U + ae^{i\delta}\mathcal{R}_\theta U\Big), \tag{6}$$

with $(\mathcal{R}_\theta U)(x, y) = U(x\cos\theta + y\sin\theta, -x\sin\theta + y\cos\theta)$. The executed twinning rotation is $\theta = \pi$, so $\mathcal{R}_\pi U(x, y) = U(-x, -y)$. The normalizer in Eq. (6) is undefined only for an exactly zero coherent field; no such case occurred, and all reported arms have finite power close to $P = 1$ W.

The registered observable is the fixed-core efficiency and its median over the propagation window:

$$C(z) = P^{-1}\int_{x^2+y^2\le r_c^2} |U_z|^2\, dA, \qquad \overline{C} = \mathrm{median}_{z\in\mathcal{Z}}\, C(z), \tag{7}$$

with core radius $r_c = 35\,\mu$m and $\mathcal{Z}$ the 21 planes from $z = 0.049$ m to 0.071 m. The comparison effect is

$$\Delta\eta = \overline{C}_{\mathrm{coherent}} - \overline{C}_{\mathrm{incoherent}}. \tag{8}$$

The overlap entering the hypothesis is

$$O = \frac{\langle U|\mathcal{R}_\theta U\rangle}{\langle U|U\rangle}, \qquad \Re O = \frac{\int \Re(U^*\mathcal{R}_\theta U)\, dA}{\int |U|^2\, dA}. \tag{9}$$

For the executed amplitude weights, the protocol labels satisfy $\alpha = 1/(1 + a^2)$: $a = 0.5$ gives $\alpha = 0.8$, $a = 1$ gives $\alpha = 0.5$, $a = 0.733799$ gives $\alpha = 0.65$, and $a = 0.333333$ gives $\alpha = 0.9$. We use this mapping only to compare with the stated law; it is not a fitted parameter.

Equation (1) is the zero-free-parameter bilinear hypothesis. It assumes that the coherent cross term scales as $2\sqrt{\alpha(1-\alpha)}\Re O$ and that the normalization and core localization do not change the sign-reversal magnitude. A minimal core-local decomposition shows why exact negation is not generic. Let $C_{\mathrm{core}} = \int_{r\le r_c} \Re(U^*\mathcal{R}_\theta U)dA$ and let η_0 be the single-arm core efficiency, with equal single-arm core integrals $\int_{r\le r_c} |U|^2 dA = \int_{r\le r_c} |\mathcal{R}_\theta U|^2 dA = \eta_0 P$. Then

$$\eta_\pm = \frac{\eta_0(1 + a^2) \pm 2a\, C_{\mathrm{core}}/P}{1 + a^2 \pm 2a\, \Re O_{\mathrm{global}}}, \tag{10}$$

where the sign is set by $\delta = 0$ or $\delta = \pi$. Equation (10) is not a new fitted law; it is the algebraic form that the stated unit-power construction implies once the core-local cross integral and the global normalization overlap are distinguished. It reduces to a bilinear-looking cross term only when the denominator and the ratio C_{core}/P are treated as phase-independent. The phase-reversal test probes precisely that assumption.

4 Numerical methods

The exploratory runs use a scalar angular-spectrum propagator with $n = 192$, square extent 2.4 mm, wavelength 633 nm, waist $300\,\mu$m, focal length 0.06 m, and power 1 W. The lens is applied, the environment is a thin lens with phase noise rms 0.02 rad and seed 20261024, and the composition fields are $\theta = \pi$, $\delta = 0$ or π, and a as listed in Table 1. The propagation window is $z = 0.049$ to 0.071 m in 21 planes. The registered core radius is $35\,\mu$m; the marginal half-width is 0.6 mm and the ridge prominence parameter is 0.25.

Table 1: Exploratory $n = 192$ angular-spectrum arms. The protocol label α is related to the stored amplitude weight a by $\alpha = 1/(1 + a^2)$. $\overline{C}_{\mathrm{coh}}$ and $\overline{C}_{\mathrm{inc}}$ are the median fixed-core efficiencies for the coherent and incoherent arms; $\Delta\eta$ is their difference. The relative phase is δ.

Case	α label	a	δ	$\overline{C}_{\mathrm{coh}}$	$\overline{C}_{\mathrm{inc}}$	$\Delta\eta$
H001	0.80	0.500000	0	0.548308	0.518068	+0.030240
H002	0.50	1.000000	0	0.552434	0.518068	+0.034366
H003	0.65	0.733799	0	0.551552	0.518068	+0.033484
H004	0.90	0.333333	0	0.543265	0.518068	+0.025198
H005	0.80	0.500000	π	0.394874	0.518068	−0.123194

The confirmation variants repeat the calibration and one held-out weight under $n = 256$, $n = 384$, a 576-point grid with 3.6 mm extent, an independent Fresnel propagator at $n = 384$, a wavelength of 640 nm, a 41-plane axial sampling, and a quantization variant. All variants use the same lens, power, composition, and core radius unless noted. The quantization variant fails the integral-convergence check and is therefore reported as a diagnostic, not as positive evidence. The confirmation records show what actually ran; a diagnostic failing a check is not converted into support for the hypothesis.

5 Results and physical interpretation

Table 1 gives the five exploratory arms. At zero relative phase, every coherent arm has a larger fixed-core efficiency than the incoherent arm. The effect is positive but not monotonic in the stored weight: it is largest at $a = 1$ ($\Delta\eta = 0.034366$), slightly smaller at $a = 0.733799$ (0.033484), and smaller again at the near-single-arm weight $a = 0.333333$ (0.025198). The calibration arm at $a = 0.5$ gives 0.030240. This is the qualitative trend expected from a coherent cross term that vanishes at the single-arm limits and is largest near equal amplitudes.

Using the calibration arm alone, and treating $\Re O$ as a common input factor (the field and rotation are identical across arms), the inferred product is

$$K\Re O = \frac{\Delta\eta(0.8)}{2\sqrt{0.8 \cdot 0.2}} = \frac{0.030240}{0.8} = 0.03780. \tag{11}$$

The corresponding predictions for the three held-out weights are 0.03780 at $\alpha = 0.5$, 0.03606 at $\alpha = 0.65$, and 0.02268 at $\alpha = 0.9$. The measured values are 0.034366, 0.033484, and 0.025198, respectively. The magnitudes of the relative deviations from these predictions are 9.1%, 7.1%, and 11.1%, all inside the pre-registered 20% band. Taken alone, the zero-phase weight dependence is therefore compatible with the bilinear scaling under the assumed mapping.

The phase-reversal arm changes the conclusion. At the same weight $a = 0.5$ and $\delta = \pi$, the coherent arm gives $\overline{C} = 0.394874$, so

$$\Delta\eta(\pi) = -0.123194. \tag{12}$$

The sign is opposite to $\Delta\eta(0) = +0.030240$, as a coherent cross-term picture would suggest, but the magnitude is about 4.07 times larger. The pre-registered exact-negation criterion is

$$\frac{|\Delta\eta(\pi) + \Delta\eta(0)|}{|\Delta\eta(0)|} = \frac{0.092954}{0.030240} = 3.07, \tag{13}$$

which exceeds the allowed 0.20 by more than an order of magnitude. The exact sign-reversal prediction is therefore rejected. A batch-level direction flag with a permissive threshold cannot override this case-specific magnitude failure.

The failure is not a small numerical blemish. Under the unit-power normalization of Eq. (6), the denominator for $\delta = 0$ is $1 + a^2 + 2a\Re O_{\mathrm{global}}$, while for $\delta = \pi$ it is $1 + a^2 - 2a\Re O_{\mathrm{global}}$. These

Table 2: Confirmation variants for the calibration arm H001 ($\alpha = 0.8$, $a = 0.5$, $\delta = 0$) and the equal-amplitude arm H002 ($\alpha = 0.5$, $a = 1$, $\delta = 0$). Entries are $\Delta\eta$. The quantization variant fails the integral-convergence check and is retained only as a diagnostic.

Condition	n	Propagator	$\Delta\eta$ H001	$\Delta\eta$ H002	Note
Exploratory	192	ASM	0.030240	0.034366	primary
Mesh256	256	ASM	0.030991	0.035220	
Mesh384	384	ASM	0.031137	0.035385	
Fresnel384	384	Fresnel	0.031136	0.035385	independent
Window576	576	ASM	0.031137	0.035385	3.6 mm extent
Wavelength640	384	ASM	0.030149	0.034330	640 nm
Axial41	384	ASM	0.024192	0.027493	41 planes
Quantization	384	ASM	0.058810	0.066600	convergence fail

denominators differ, and the fixed-core integral of the cross term is C_{core}, not the global overlap $\Re O_{\mathrm{global}}$. Equation (10) shows that an exact negation of $\Delta\eta$ would require both the denominator and the core-local ratio C_{core}/P to be phase-independent. Neither condition is implied by the stated construction. The observed sign reversal with a larger negative magnitude is qualitatively consistent with a smaller denominator at $\delta = \pi$ and a core-local cross integral that is not equal to the global overlap. The present data do not log $\Re O$ or C_{core}, so this interpretation is a constraint on the law, not a fitted explanation.

Table 2 shows that the calibration difference is stable near 0.031 for the $n = 256$, $n = 384$, Fresnel, and larger-window variants, with a small decrease at 640 nm and a larger decrease under 41-plane axial sampling. The equal-amplitude difference is similarly stable near 0.035 for the main variants. The quantization variant gives a roughly twice larger apparent difference and fails the integral-convergence check; it is not used as positive evidence. Absolute $\overline{C}$ is more grid-sensitive than $\Delta\eta$: for H001 the coherent value is 0.5483 at $n = 192$, 0.6118 at $n = 256$, and 0.5952 at $n = 384$. This grid dependence is a calibration issue for absolute concentration, but the difference between coherent and incoherent arms is comparatively stable across the non-quantized confirmation runs.

The optimized classical controls provide context for the absolute observable. Under the same geometry and $n = 192$ ASM, a defocus control ($q_2 = -0.30$) reaches $\overline{C} = 0.599224$, a quadratic–quartic control ($q_2 = -1.2$, $q_4 = 0.35$) reaches 0.608637, and a four-parameter control ($q_2 = -1.148895$, $q_3 = -0.006480$, $q_4 = 0.345100$, ripple -0.021287) reaches 0.619423. These are higher than the coherent Pearcey arm at 0.548308. The controls were optimized for the same absolute core efficiency, so they show that the twinning arm is not a competitive focusing design in this geometry. They are not used to define the twinning gain, and they are not interpreted as a negated optimizer score or as a claim about absolute concentration in other geometries.

The caustic-specificity question is not answered by these runs. The required parity-matched non-centrosymmetric control, with matched odd fraction and a comparison of the core-local coefficient, was not executed. No random-phase Gaussian genericity arm, no f_{odd} matching, and no K_{gen} comparison are present. The observation that a Pearcey input shows a coherent cross-term effect therefore cannot be promoted to caustic specificity. A generic coherent cross-term law remains a live alternative, and the present evidence does not distinguish it from a caustic-specific one.

Figure 1 collects the executed summary used for the tables. The quantitative statements in this paper are taken from the registry values in Tables 1 and 2; the figure is a diagnostic summary, not an independent measurement.

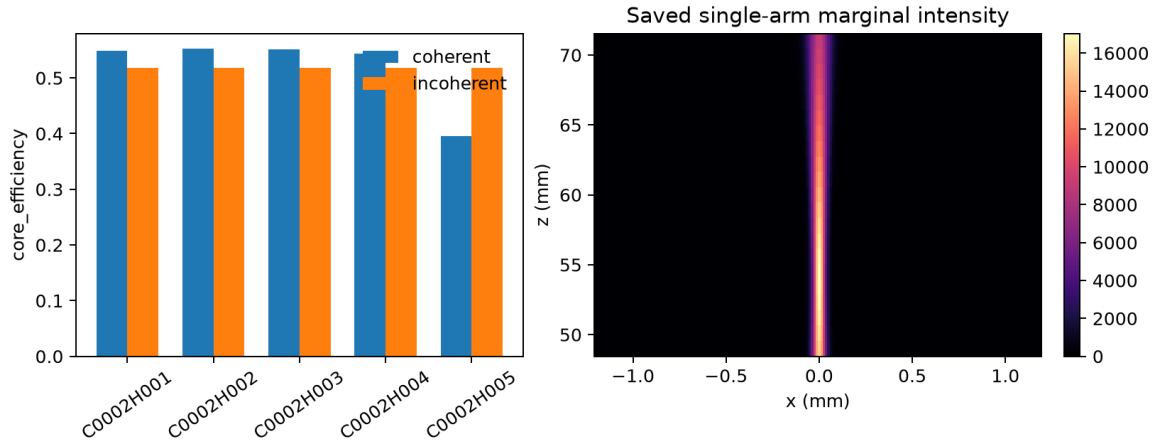


Figure 1: Summary of the executed numerical runs reported in Tables 1 and 2. The figure is generated from the same registry values and is used as a visual diagnostic; numerical claims in the text are based on the tabulated values.

6 Conclusions

In the frozen ideal-lens geometry studied here, coherent superposition with a 180° rotated copy changes the fixed-core efficiency of a Pearcey-type input. At zero relative phase the change is positive for all executed weights, largest near equal amplitudes, and small near the single-arm limit. Under the inferred mapping $\alpha = 1/(1 + a^2)$, the three held-out exploratory weights deviate in magnitude by 9.1%, 7.1%, and 11.1% from the calibration prediction using $K\Re O = 0.03780$. That is a suggestive weight trend, but it is not sufficient to establish the proposed zero-free-parameter bilinear law.

The decisive phase-reversal test fails. At the calibration weight, $\Delta\eta(0) = +0.030240$ and $\Delta\eta(\pi) = -0.123194$. The sign reverses, but the magnitude does not: the sum ratio is 3.07, far above the allowed 0.20. Exact negation is not algebraically implied by the unit-power normalization, because the normalization denominator changes with relative phase and the core-local cross integral is not the global overlap. The proposed global-overlap law is therefore not supported by the executed evidence.

The caustic-specificity claim is also unsupported. The matched non-centrosymmetric genericity control was not executed, so the data cannot distinguish a caustic-specific coefficient from generic coherent cross-term shaping. The optimized classical controls reach higher core efficiency than the coherent Pearcey arm, which rules out a competitive focusing interpretation but does not by itself test the cross-term mechanism. The overall result is a bounded negative for the stated law and a bounded positive for the existence of phase- and weight-dependent coherent cross-term shaping. A decisive follow-up would log $\Re O$, C_{core}, and the normalization denominators, test the core-local form of Eq. (10) at both relative phases, execute the centrosymmetric nulls, and run the parity-matched genericity arm with frozen thresholds.

Disclosures

Auto BeamScientist adapts Sakana AI Scientist-v2 [4]. The author and PDF author are Auto BeamScientist Project. This manuscript reports numerical simulations; no physical experiment was performed. The figure and tables are generated from the executed registry values.

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